

KHEYDZHAN, Russian FederationWMO: **310830**Lat: **59.967N**Lon: **144.783E**Elev: **121**StdP: **99.88**Time Zone: **10.00 (E10)**Period: **86-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.5	-42.4	-46.2	0.0	-44.3	-44.3	0.0	-42.4	4.2	-14.2	3.5	-15.2	0.9	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	26.0	16.6	24.0	15.8	21.9	15.0	17.6	23.8	16.6	22.5	15.6	20.6	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	11.0	19.4	14.2	10.2	17.7	13.4	9.7	16.9	50.3	23.7	47.1	22.8	44.1	20.8	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.2	3.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.8	-26.8	-22.4	-15.6	-4.7	3.6	11.4	14.1	12.9	5.8	-5.8	-18.6	-24.9
	DBStd	16.05	8.93	7.78	6.61	4.52	3.17	3.23	2.49	3.12	4.14	6.87	8.89	9.66
	HDD10.0	6082	1140	908	794	442	198	22	1	11	136	491	857	1082
	HDD18.3	8821	1398	1142	1053	692	456	207	132	170	375	749	1107	1341
	CDD10.0	305	0	0	0	0	1	65	129	100	11	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	226	0	0	0	0	0	79	108	39	0	0	0	0
	CDH26.7	33	0	0	0	0	0	11	20	2	0	0	0	0
Wind	WSAvg	1.2	0.8	1.0	1.1	1.3	1.7	1.7	1.3	1.1	1.0	1.0	1.0	1.0
Precipitation	PrecAvg	704	26	12	23	36	45	61	92	121	103	90	65	31
	PrecMax	1030	80	50	85	96	126	110	226	295	228	171	282	163
	PrecMin	349	1	2	2	2	7	13	20	46	38	12	13	1
	PrecStd	161	21	10	21	22	25	27	46	64	57	48	58	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.4	-2.0	1.5	9.0	19.1	28.7	28.8	26.5	20.9	10.0	0.7	-3.9
		MCWB	-5.1	-3.4	-2.1	3.2	10.3	16.6	19.6	16.8	13.8	4.4	0.0	-4.8
	2%	DB	-7.6	-4.1	-1.0	6.5	16.4	25.7	26.2	24.4	18.4	6.9	-1.5	-5.4
		MCWB	-8.7	-5.1	-3.6	1.2	8.5	15.5	17.5	16.5	12.0	3.5	-2.1	-6.1
	5%	DB	-10.3	-7.0	-2.6	4.8	14.3	23.5	23.8	22.6	16.3	4.8	-3.0	-7.3
		MCWB	-11.2	-8.2	-5.2	0.3	7.7	14.2	16.7	15.7	11.3	2.3	-3.9	-8.4
	10%	DB	-12.5	-10.0	-4.2	3.1	12.0	21.2	21.4	20.4	14.0	3.2	-4.7	-10.1
		MCWB	-13.3	-11.4	-6.3	-0.6	6.3	13.3	15.7	14.6	10.1	1.5	-5.4	-11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.2	-2.8	-1.6	3.6	11.0	17.8	20.8	18.6	14.7	5.5	0.7	-4.6
		MCDB	-4.6	-2.3	0.9	7.9	17.8	24.5	27.8	24.1	19.6	8.1	1.3	-4.3
	2%	WB	-8.8	-5.0	-3.7	1.5	9.2	16.2	18.3	17.1	13.6	4.2	-2.0	-6.0
		MCDB	-7.8	-4.0	-1.1	5.1	15.4	24.2	24.4	22.6	16.7	5.9	-1.6	-5.3
	5%	WB	-11.1	-8.2	-5.0	0.6	7.8	15.0	17.0	16.2	12.0	2.9	-3.7	-8.6
		MCDB	-10.2	-6.8	-2.9	3.7	13.8	22.2	23.0	21.2	15.2	4.5	-3.3	-7.6
	10%	WB	-13.3	-11.4	-6.2	-0.1	6.5	13.7	15.9	15.2	10.5	1.6	-5.3	-11.5
		MCDB	-12.4	-10.0	-4.2	2.6	11.5	20.4	20.8	19.4	13.6	2.9	-4.6	-10.3
Mean Daily Temperature Range	5% DB	MDBR	10.9	16.0	16.9	13.3	12.0	14.9	10.5	12.6	12.6	11.6	11.3	10.2
		MCDBR	8.8	11.6	17.4	15.5	18.0	22.3	19.1	18.6	16.8	11.6	6.7	7.8
		MCWBR	8.2	10.6	14.1	10.6	10.6	12.2	10.8	11.0	10.5	8.1	6.5	7.5
	5% WB	MCDBR	8.8	9.7	14.8	12.5	17.1	20.1	16.7	15.2	14.1	9.8	7.0	8.1
		MCWBR	8.2	9.0	12.3	8.4	10.2	11.2	10.0	9.3	9.1	7.2	6.8	7.7
Clear-Sky Solar Irradiance	taub	0.229	0.273	0.334	0.454	0.467	0.430	0.483	0.431	0.347	0.311	0.250	0.209	
	taud	2.040	2.071	1.951	1.692	1.852	2.202	2.069	2.212	2.413	2.192	2.035	1.955	
	Ebn at Noon	544	714	764	716	751	801	739	753	776	657	488	403	
	Edh at Noon	45	79	125	197	181	130	146	117	81	70	45	32	
All-Sky Solar Radiation	RadAvg	0.46	1.46	3.24	5.04	5.32	5.28	4.62	3.75	2.68	1.54	0.63	0.24	
	RadStd	0.04	0.08	0.13	0.39	0.50	0.48	0.70	0.43	0.36	0.17	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page